

Quanhao Sun

☎ 213-561-9627 | ✉ quanhaos@usc.edu | 🔗 linkedin.com/in/quanhao-sun | 🌐 quanhaos.github.io

EDUCATION BACKGROUND

University of Southern California

Aug. 2022 – Present

PhD in Biomedical Engineering

Los Angeles, CA

- **Lab:** Laboratory of Neuro Imaging (LONI); Advisors: Prof. Hosung Kim, Prof. Michael Khoo
- **Coursework:** Machine Learning, Digital Signal Processing, Magnetic Resonance Imaging and Reconstruction
- **Research:** Generative AI for Learning Meaningful Medical Imaging Representations under Limited Ground Truth

Southeast University

Aug. 2018 – Jun. 2022

B.E. in Biomedical Engineering

Nanjing, China

- **Advisors:** Peng Xie; Yu Sun; Zongcai Ruan; Yu Zhang
- **Coursework:** Medical Image Processing, Medical Imaging Principles, Fundamentals of Data Structures

SKILLS

Programming Languages : Python, C++, MATLAB, JavaScript, HTML/CSS, SQL, R

Tools : PyTorch, PyTorch Lightning, TensorFlow, nnU-Net, ANTs, MONAI, nibabel, PyQt6, Git, L^AT_EX, Vaa3D

PROJECTS

Generative Synthesis of Black-Blood MRI from T2-FLAIR Enables Contrast-free Quantitative Assessment of the Parasagittal Dural Space and Glymphatic Function

- Developed a cross-contrast synthesis pipeline generating Black Blood MRI from FLAIR via a two-stage 3D VQGAN + modality-conditional MaskGIT framework, with a VQ tokenizer trained on mixed BB/FLAIR volumes.
- Adapted MaskGIT to predict masked BB tokens conditioned on FLAIR tokens, using iterative parallel decoding with 2-phase cosine-sine combined mask scheduling and confidence-based refinement for full-volume synthesis.
- Built BB-to-FLAIR registration, isotropic resampling, and sliding-window NIfTI inference; integrated ROI-aware evaluation around parasagittal dural space to support PSD volume quantification and biomarker analyses.

Correction of Motion-Affected MRI Images via Motion-Adaptive Diffusion Model

- Developed a Motion-Adaptive Diffusion Model (MADM) for 3D MRI motion correction in a VQGAN latent space, using an age-conditioned, attention-enhanced 3D U-Net denoiser with Curvelet/wavelet decomposition and consistency losses to preserve fine anatomy and tissue contrast.
- Pretrained VQGAN on over 10k clean adult and infant MRI volumes, then fine-tuned with motion-clean pairs; trained MADM via paired supervised learning plus unpaired data adaptation for cross-population generalization.
- Validated with 5-fold cross-validation using PSNR, SSIM, NIQE, and GMSD; demonstrated motion correction on adult and neonatal cohorts (ISMRM/BMES 2025).

Baby MRI Brain Development Biomarker Arc Segmentation

- Built a clinician-in-the-loop 3D nnU-Net v2 pipeline for arc segmentation in fetal/infant T2 MRI: trained Model A on 20 manual labels, applied it to 73 new cases, and retrained Model C on clinician-corrected labels.
- Compared automatic segmentation with manual labels using Dice, IoU, HD95, volume, Bland–Altman analyses, and F-tests under 5-fold cross-validation.
- Applied N4 bias-field correction as image preprocessing and registered MRI and predicted masks to the STA38exp fetal atlas via ANTs rigid+SyN transforms for cross-subject comparison.

Regional Brain Age Prediction with Graph Convolutional Networks from Multi-modal MRI

- Developed a Laplacian-depth white-matter surface reconstruction method with U-Net guidance and an improved ODE contraction model to extract multi-layer WM intensity features from T1/T2-FLAIR MRI.
- Built a GCN pipeline combining cortical GM features and depth-stratified WM ROIs to predict regional brain age and Brain Age Gap (BAG) on ENIGMA and population cohorts, evaluated against chronological age.

Multi-modal Physiological Signal Analysis Software for EDF

- Built a desktop and CLI tool for EDF physiological signal analysis (e.g., PPG) with filtering, CWT/STFT visualization, feature extraction, and session save/load.
- Designed a shared analysis engine with lazy window loading, caching, and batch PNG/CSV/JSON export, covered by automated end-to-end tests.

Accelerated MRI Reconstruction with Cascaded Multi-Scale Wavelet Networks from Undersampled K-space

- Built a cascaded multi-scale wavelet reconstruction network with iterative refinement to recover high-quality MRI from undersampled k-space.
- Benchmarked against variational networks, compressed sensing, and parallel imaging.

Brain Tumor Segmentation for Multi-Contrast MRI Based on Deep Learning

- Applied FuseNet to segment brain tumors from multi-contrast MRI (T1, T2, DWI, FLAIR) with an information-selection step to identify major contributing modes.

Neural Functional Connectivity Brain Atlas Database

- Built a cortex–thalamus functional connectivity atlas with 3D visualization and clustering of hierarchical brain regions.

Thyroid Nodule Segmentation for Ultrasound Images Based on Deep Learning

- Designed a dual-pathway high-resolution encoder–decoder with a convolutional block attention module and improved segmentation of noisy, boundary-discontinuous thyroid nodules.

Vaa3D Neuron Image Enhancement Plugin Development

- Developed and maintained Vaa3D plugins for gigabyte-scale neuron visualization and VR-assisted observation.

Mesoporous Prussian Blue Nanoparticles for Acute Myeloid Leukemia Treatment

- Modified Zwitterionic and CXCR4 antagonist peptide E5 onto nanoparticle surfaces, yielding photothermal effect, reduced nonspecific protein adsorption, targeting capability, and biocompatibility.

PUBLICATIONS

- Chai Y, Zhang H, Kim AS, **Sun Q**, Chen S, Ha J, Kang KW, Cen H, et al. Multimodal Free-Water Imaging Links Cardiometabolic Risk to Periarterial Dysfunction and Amyloid Accumulation in Early Alzheimer's, RESEARCH SQUARE. 2026.
- Coloma M, Poddar A, **Sun Q**, Xu D, Calabrese E, Glass HC, Wu YW, Kim H, Paredes M, Li Y. Association between Perinatal Risk Factors and MRI Injury with Arc Volume in Neonates with Hypoxic Ischemic Encephalopathy, AMERICAN SOCIETY OF PEDIATRIC NEURORADIOLOGY. 2026.
- Poddar A, Coloma M, **Sun Q**, Xu D, Calabrese E, Glass HC, Wu YW, Kim H, et al. Using in vivo MR Imaging as a Biomarker for the Expanded Neonatal Subventricular Zone, AMERICAN SOCIETY OF PEDIATRIC NEURORADIOLOGY. 2026.
- **Sun Q**, Chai Y, Li S, Duan X, Khoo M, Kim H. Latent Diffusion Model to Remove Motion Artifacts in Neonatal MRI, BIOMEDICAL ENGINEERING SOCIETY. 2025.
- **Sun Q**, Chai Y, Khoo MCK, Kim H. Correction of Motion-Affected MRI Images via Motion-Adaptive Diffusion Model, INTERNATIONAL SOCIETY FOR MAGNETIC RESONANCE IN MEDICINE, Abstract #3980. 2024.
- Bai H, **Sun Q**, Kong F, et al. Zwitterion-Functionalized Hollow Mesoporous Prussian Blue Nanoparticles for Targeted and Synergetic Chemo-Photothermal Treatment of Acute Myeloid Leukemia, JOURNAL OF MATERIALS CHEMISTRY B. May 2021.

AWARDS

- Meritorious Winner of 2021 Mathematical Contest In Modeling
- 1st Prize of 2020 Contemporary Undergraduate Mathematical Contest In Modeling
- 3rd Prize of 2020 National Biomedical Engineering Innovation Design Competition (top undergraduate award)
- Individual Gold Medal and Team Gold Medal of 2012 World Mathematics Team Championship Junior Group